**UNIVERSITI PUTRA MALAYSIA**

**FAKULTI SAINS KOMPUTER DAN TEKNOLOGI MAKLUMAT**

CCS3700 – EMBEDDED SYSTEM PROGRAMMING

LAB 3

# POTENTIOMETER AND FLICKERS

LECTURER'S NAME:

DR. KHAIRUL AZHAR

STUDENT'S NAME:

MUHAMMAD ASLAM ROFI BIN MOHD RASU

MATRIC NO:

230405

SUBMITTED DATE:

24/4/2026

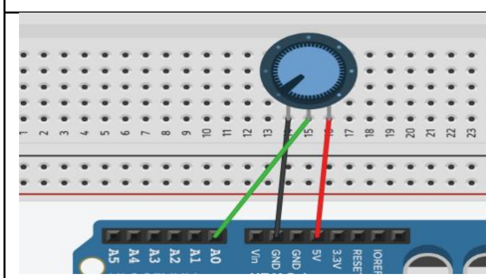
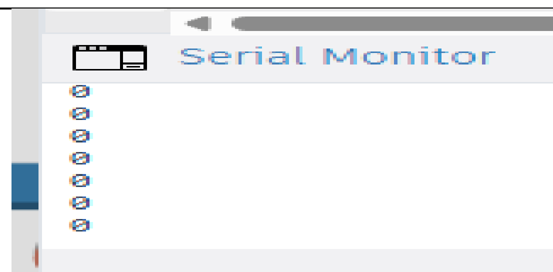
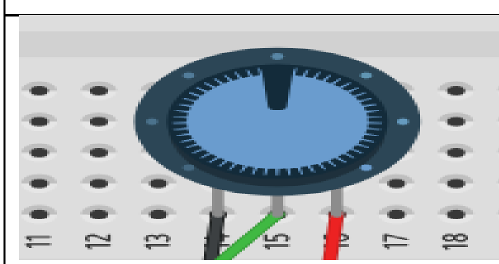
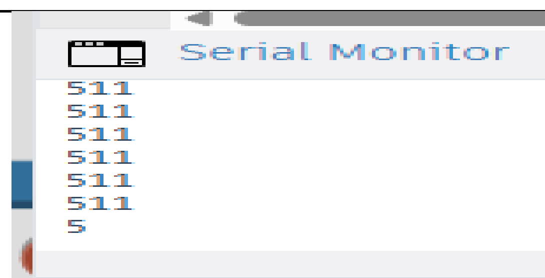
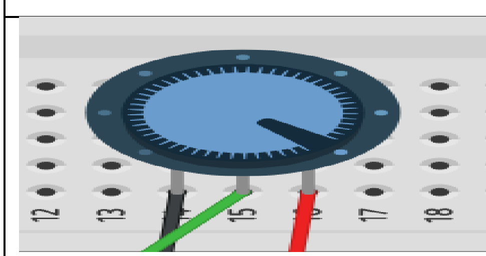
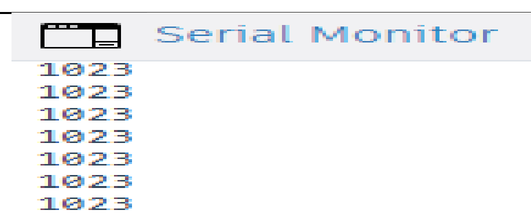
## Part 1 Analog Read Serial

A potentiometer measuring instrument is essentially a voltage divider used for measuring electric potential (voltage). Potentiometers are commonly used to control electrical devices such as volume controls on audio equipment.

```
void setup() {
  // Begin serial communications at 9600 bits of data per second [cite: 42, 43]
  Serial.begin(9600);
}

void loop() {
  // Store the resistance value from the potentiometer in a variable [cite: 44, 45]
  int sensorValue = analogRead(A0);

  // Print this information to your serial window as a decimal (DEC) value [cite: 46, 47, 48]
  Serial.println(sensorValue, DEC);

  // Adding a tiny delay makes the Tinkercad simulation run much smoother
  delay(10);
}
```

## Instruction

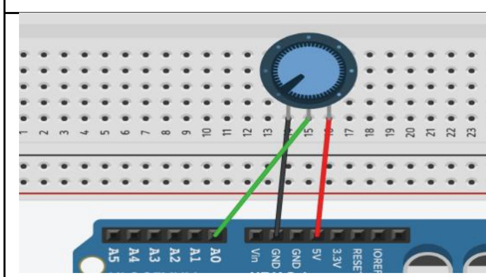
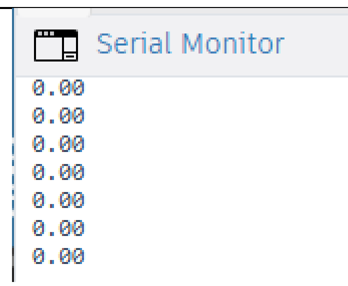
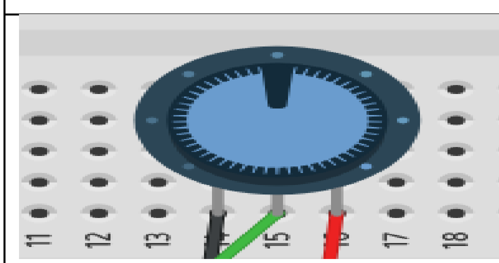
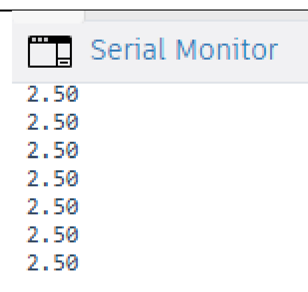
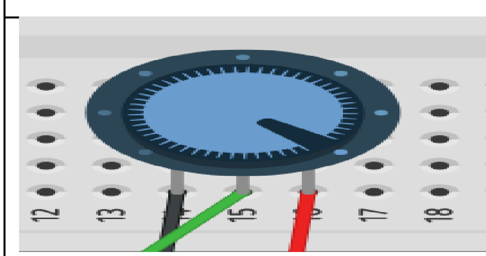
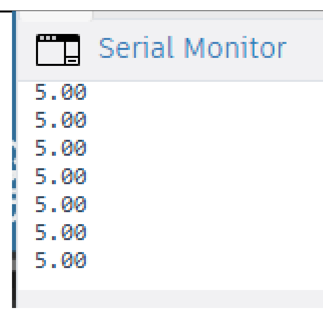
Write the required program and compile it into your Arduino board. In the program, the only thing that you will do in the setup function is to begin serial communications, at 9600 bits of data per second, between your Arduino and your computer.

| POTENTIOMETER | SERIAL MONITOR |
| --- | --- |
|  |   |
|  |   |
|  |   |

## **Part 2 Read Analog Voltage**

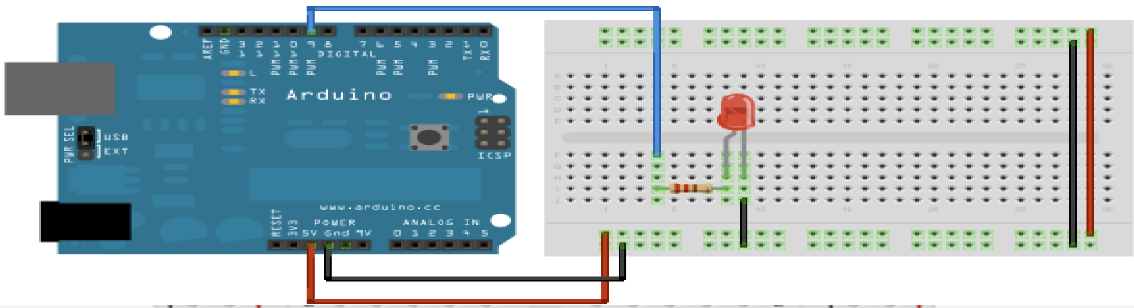
In this lab exercise, students will attempt to read an analog input on pin 0, convert the values from *analogRead()* into voltage, and print them out to the serial monitor.

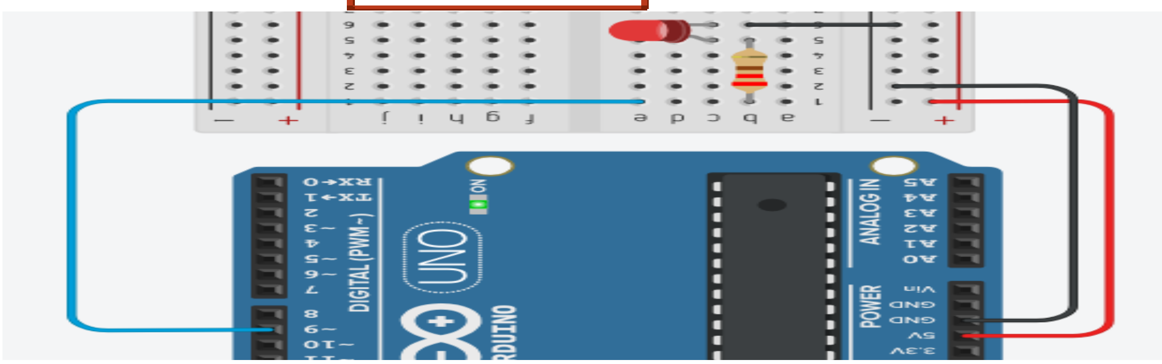
```
void setup() {
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(A0);
  float voltage = sensorValue * (5.0 / 1023.0);
  Serial.println(voltage);
  delay(10);
}
```

<table>
<tr><th>POTENTIOMETER</th><th>SERIAL MONITOR</th></tr>
<tr><td></td><td>
</td></tr>
<tr><td>
</td><td>
</td></tr>
<tr><td>
</td><td>
</td></tr>
</table>

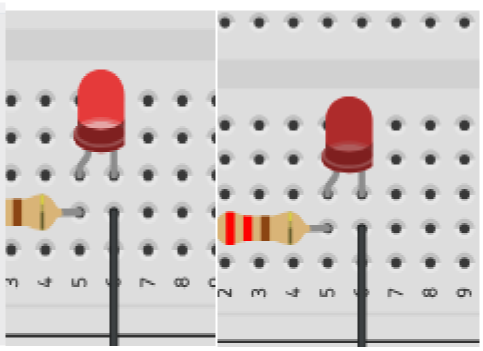
## Part 3 Fading

Demonstrates the use of the ***analogWrite()*** function in fading an LED off and on. analogWrite uses pulse width modulation (PWM), turning a digital pin on and off very quickly, to create a fading effect.




```
int ledPin = 9;
int brightness = 0;
int fadeAmount = 5;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  analogWrite(ledPin, brightness);

  brightness = brightness + fadeAmount;

  if (brightness <= 0 || brightness >= 255) {
    fadeAmount = -fadeAmount;
  }

  delay(30);
}
```



This code makes it flickers ever 3 seconds or so. We can also change the code if wanted for higher or lower delay

**END OF LAB**